

Assignment 1: Array Creation and Manipulation

1. Create a NumPy array of shape (5, 5) filled with random integers between 1 and 20. Replace all the elements in the third column with 1.
2. Create a NumPy array of shape (4, 4) with values from 1 to 16. Replace the diagonal elements with 0.

Assignment 2: Array Indexing and Slicing

1. Create a NumPy array of shape (6, 6) with values from 1 to 36. Extract the sub-array consisting of the 3rd to 5th rows and 2nd to 4th columns.
2. Create a NumPy array of shape (5, 5) with random integers. Extract the elements on the border.

Assignment 3: Array Operations

1. Create two NumPy arrays of shape (3, 4) filled with random integers. Perform element-wise addition, subtraction, multiplication, and division.
2. Create a NumPy array of shape (4, 4) with values from 1 to 16. Compute the row-wise and column-wise sum.

Assignment 4: Statistical Operations

1. Create a NumPy array of shape (5, 5) filled with random integers.
Compute the mean, median, standard deviation, and variance of the array.
2. Create a NumPy array of shape (3, 3) with values from 1 to 9.
Normalize the array so that the values have a mean of 0 and a standard deviation of 1.

Assignment 5: Broadcasting

1. Create a NumPy array of shape (3, 3) filled with random integers.
Add a 1D array of shape (3,) to each row of the 2D array using broadcasting.
2. Create a NumPy array of shape (4, 4) filled with random integers.
Subtract a 1D array of shape (4,) from each column of the 2D array using broadcasting.

Assignment 6: Linear Algebra

1. Create a NumPy array of shape (3, 3) representing a matrix.
Compute its determinant, inverse, and eigenvalues.
2. Create two NumPy arrays of shape (2, 3) and (3, 2). Perform matrix multiplication on these arrays.

Assignment 7: Advanced Array Manipulation

1. Create a NumPy array of shape (3, 3) with values from 1 to 9.
Reshape the array to shape (1, 9) and then to shape (9, 1).
2. Create a NumPy array of shape (5, 5) filled with random integers.
Flatten the array and then reshape it back to (5, 5).

Assignment 8: Fancy Indexing and Boolean Indexing

1. Create a NumPy array of shape (5, 5) filled with random integers. Use fancy indexing to extract the elements at the corners of the array.
2. Create a NumPy array of shape (4, 4) filled with random integers. Use boolean indexing to set all elements greater than 10 to 10.

Assignment 9: Structured Arrays

1. Create a structured array with fields name, age, and weight. Add some data and sort the array by age.
2. Create a structured array with fields x and y, both as integers. Add some data and compute the Euclidean distance between each pair of points.

Assignment 10: Masked Arrays

1. Create a masked array of shape (4, 4) with random integers and mask the elements greater than 10. Compute the sum of the unmasked elements.
2. Create a masked array of shape (3, 3) with random integers and mask the diagonal elements. Replace the masked elements with the mean of the unmasked elements.